NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES ISLAMABAD

CS-1002 PROGRAMMING FUNDAMENTALS SPRING 2026

ASSIGNMENT-01

Instructions:

Dear students' failure to submit according to the below format would result in zero marks in the relevant evaluation.

i. For each question in your assignment, make a separate cpp file e.g. for question 1, make ROLL-NUM_SECTION_Q#.cpp (24i-0001_Q1.cpp) and so on. Each file that you submit must contain your name, student-id, and assignment # on top of the file in comments.
ii. Combine all your work in one folder. The folder must contain only .cpp files (no binaries, no exe files etc.).
iii. Run and test your program on a lab machine before submission.
iv. Rename the folder as ROLL-NUM_SECTION (e.g. 24i-0001) and compress the folder as a zip file. (e.g. 24i-0001.zip). do not submit .rar file.
v. Submit the .zip file on Google Classroom within the deadline.
vi. Submission other than Google classroom (e.g. email etc.) will not be accepted.
vii. The student is solely responsible to check the final zip files for issues like corrupt file, virus in the file, mistakenly exe sent. If we cannot download the file from Google classroom due to any reason it will lead to zero marks in the assignment.
viii. Displayed output should be well mannered and well presented. Use appropriate comments and indentation in your source code.
ix. Total Marks: 65.
x. If there is a syntax error in code, zero marks will be awarded in that part of assignment.
xi. Your code must be generic.

Late Submission policy will be applied as described in course outline.

Plagiarism: Plagiarism is not allowed. If found plagiarized, you will be awarded zero marks in the assignment (copying from the internet is the easiest way to get caught).

Note: Follow the given instruction to the letter, failing to do so will result in a zero.

Deadline to submit assignment is 20st February 2026 11:59 PM.

Question 01 (Marks 05)

Write a C++ program which accepts a Uppercase alphabet as input and convert it to lower case alphabet.

Example 01:

Input: A

Output: a

Example 02:

Input: Z

Output: z

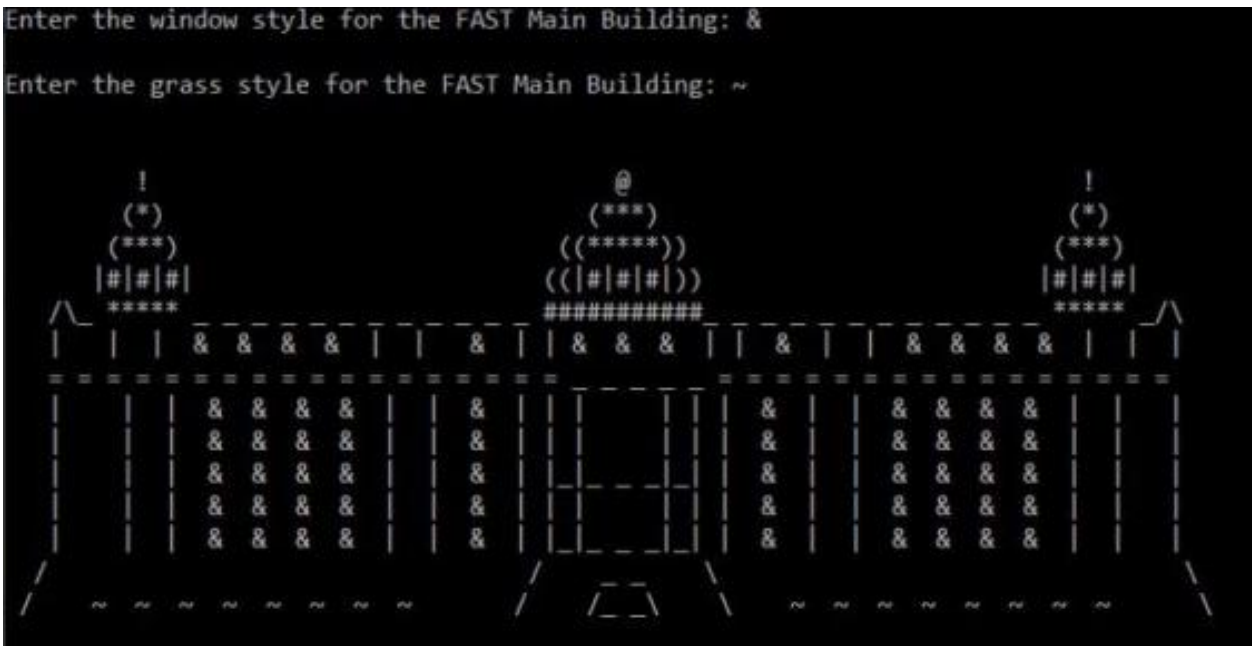
Question 02 (Marks 05)

Write a program that prints the following using cout statements.

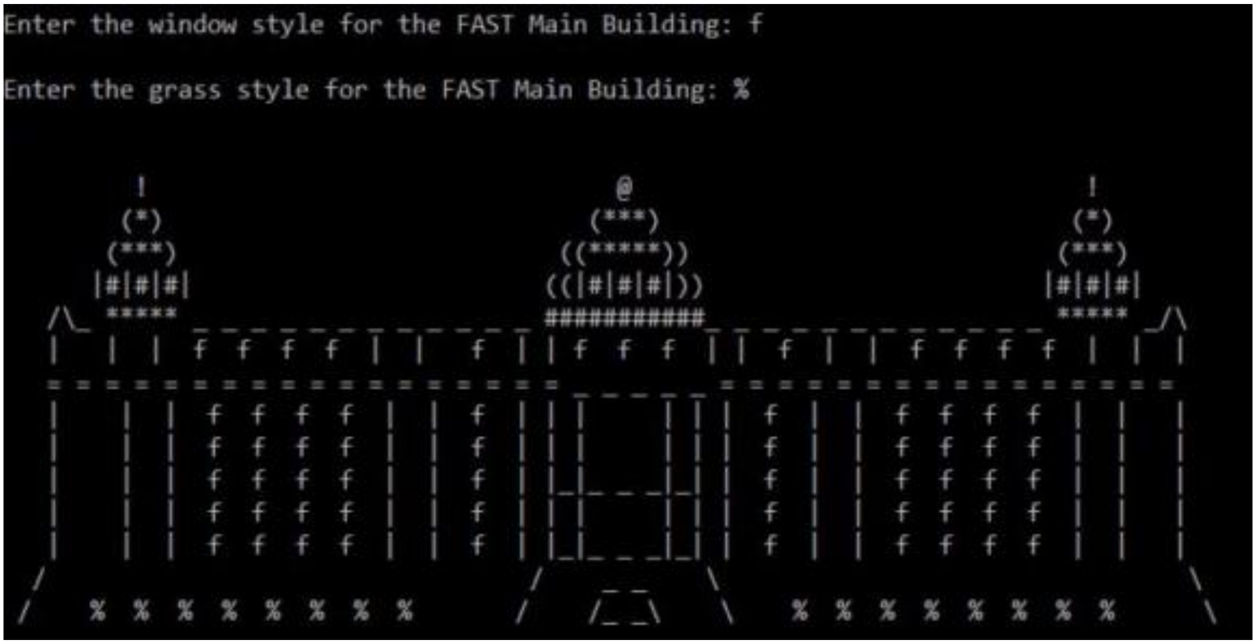
Question 03 (Marks 10)

Write a C++ program that outputs **FAST Main Building pattern** using I/O manipulator <iomanip> appropriately *(i.e. setw functons).* The output will vary based on two inputs that the user will provide: window style and grass style. By simply changing the input characters, the output will change (somewhat) dramatically. In the end, your program must match the outputs shown for the give characters for window and grass style EXACTLY. *(Note: You cannot use any string literals containing spaces in cout.* ***And use single cout statment****)*

Here is the sample program output with an input character & for window style and ~ for grass style:



Here is the sample program output with an input character f for window style and % for grass style:



Question 04 (Marks 10)

Write a C++ program to calculate the given formula below. The value of i, a and b needs to be input by user while u = 3 and p = 4 (all the variable's data type must be int except result (double)).

$$\text{result} = \frac{\sqrt{\mu(i)^{\frac{a}{b}}(i^2-1)}}{\sqrt{\rho(i)-2}+\sqrt{\rho(i)-1}}$$

Question 05 (Marks 05)

An airport currency exchange counter receives an amount in **Pakistani Rupees (PKR)** from a traveler (integer input within the range Rs. 500 to Rs. 200000).

Before returning the amount:

1. The counter deducts a **Service Charge of 3%** of the total amount.
2. The remaining amount is returned using the **minimum number of available currency denominations**.

Available denominations:

- Rs. 5000
- Rs. 1000
- Rs. 500
- Rs. 100
- Rs. 50
- Rs. 10
- Rs. 1

Your program must:

- Take the total amount as input
- Calculate the service charge
- Compute the remaining payable amount
- Display the number of each denomination required

⚠ No loops, no conditionals, no built-in functions allowed.

Example

Input:
Amount: 18750

Output:

Service Charge (3%): 562
Remaining Amount: 18188

Denomination : Count

5000 : 3

1000 : 3

500 : 0

100 : 1

50 : 1

10 : 3

1 : 8

Question 06 (Marks 10)

To save disk space Time field in the directory entry is 2 bytes long. Distribution of different bits which account for hours, minutes and seconds is given below:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
H	H	H	H	M	M	M	M	M	M	S	S	S	S	S	S

Write a C++ Program that take input two-byte time entry and appropriately separates hours, minutes and seconds using suitable bitwise operators. Think cleverly. The program executes in following manner:

Example 01:

Enter a two-byte Time Value: 8849

Time is 2 hrs 10 mins 17 secs

Example 02:

Enter a two-byte Time Value: 4879

Time is 1 hrs 12 mins 15 secs

(Marks 10) Question 08: Write a program, which takes two 8-bit integers value and add them. Remember, you can only use bitwise (& | ^ ~ etc.) or assignment (=) operators for this problem.

Example01:

Enter number 1: 25

Enter number 2: 63

Output: 88

Example02:

Enter number 1: 15

Enter number 2: 7

Output: 22

Question 07 (Marks 20)

Write a C++ program for the Tax Advisory Portal that helps Pakistani citizens calculate their monthly income tax liability. The tax calculation follows the Pakistani income tax system for both salaried and non-salaried individuals for the fiscal year 2023-2024. The program also considers a 25% tax rebate for teachers employed by non-profit educational institutions.

- The user will input their **annual income** in Pakistani Rupees (PKR) and specify whether they are **salaried or non-salaried**.
- The user must also specify whether they are a **teacher working in a non-profit organization**
- Based on the input, the program will calculate the **annual income tax** using the tax brackets below.
- If the user is a teacher in a non-profit organization, the program will apply a **25% rebate** on the calculated annual tax.
- The program will output the **monthly tax deduction**, assuming the annual tax is divided equally over 12 months.
- If the annual income is less than PKR 600,000, the program should inform the user that no tax is deducted.

Tax Brackets for Salaried Individuals (2023-2024):

1. **Up to PKR 600,000**: No tax.

2. **PKR 600,001 to PKR 1,200,000**: 2.5% of the amount exceeding PKR 600,000. 3. **PKR 1,200,001 to PKR 2,400,000**: PKR 15,000 + 12.5% of the amount exceeding PKR 1,200,000.

4. **PKR 2,400,001 to PKR 3,600,000**: PKR 165,000 + 20% of the amount exceeding PKR 2,400,000.

5. **PKR 3,600,001 to PKR 6,000,000**: PKR 405,000 + 25% of the amount exceeding PKR 3,600,000.

6. **Above PKR 6,000,001**: PKR 1,005,000 + 32.5% of the amount exceeding PKR 6,000,000.

Tax Brackets for Non-Salaried Individuals (2023-2024):

1. **Up to PKR 600,000**: No tax.
2. **PKR 600,001 to PKR 1,200,000**: 5% of the amount exceeding PKR 600,000. 3. **PKR 1,200,001 to PKR 2,400,000**: PKR 30,000 + 12.5% of the amount exceeding PKR 1,200,000.
3. **PKR 2,400,001 to PKR 3,600,000**: PKR 180,000 + 17.5% of the amount exceeding PKR 2,400,000.
4. **PKR 3,600,001 to PKR 6,000,000**: PKR 390,000 + 22.5% of the amount exceeding PKR 3,600,000.
5. **Above PKR 6,000,001**: PKR 975,000 + 30% of the amount exceeding PKR 6,000,000.

Input:

- The program should prompt the user to enter their **annual income** in Pakistani Rupees (PKR).
- The user will specify whether they are **salaried** or **non-salaried**.
- The user will specify whether they are a **teacher in a non-profit organization** by entering 'yes' or 'no'.

Output:

• The program will display:

1. The **annual income tax** based on the appropriate tax brackets (salaried or non salaried).
2. If the user is a teacher in a non-profit, apply the **25% tax rebate** on the calculated annual tax.
3. The **monthly tax deduction** (annual tax after rebate divided by 12).

Example Scenario 1:

A **salaried employee** earns an annual salary of PKR 1,500,000 and is **not a teacher**. The program will calculate the tax based on the salaried brackets and display the monthly tax deduction.

Example Scenario 2:

A **non-salaried individual** earns PKR 2,000,000 annually and is a **teacher in a non-profit**. The program will calculate the tax using the non-salaried brackets, apply the 25% rebate, and then display the monthly tax deduction based on the reduced tax.

Constraints:

- The program must handle tax calculation for both salaried and non-salaried individuals.
- You **must use if-else** statements to determine tax brackets and rebates.
- **No loops** should be used in the solution.
- The program should perform input validation to ensure that the income entered is a positive number and correctly validate whether the user is salaried and/or a teacher in a non-profit.